Lab-2:

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
from nltk.corpus import wordnet

# Download required NLTK data resources
nltk.download('punkt', quiet=True)
nltk.download('punkt_tab', quiet=True)
nltk.download('stopwords', quiet=True)
nltk.download('wordnet', quiet=True)
nltk.download('averaged_perceptron_tagger', quiet=True)
nltk.download('averaged_perceptron_tagger_eng', quiet=True)

# Input text
text = ("The students are studying different subjects and they are enjoying "
        "their studies. The teachers are teaching the subjects very
effectively.")

# 1. Tokenization (alphanumeric tokens only to remove punctuation)
tokens = [w for w in word_tokenize(text) if w.isalnum()]

# 2. Stop-word Removal (preserving case-lowered comparison)
stop_words = set(stopwords.words('english'))
filtered_tokens = [w for w in tokens if w.lower() not in stop_words]

# 3. Stemming (using Porter Stemmer)
stemmer = PorterStemmer()
stemmed_words = [stemmer.stem(w) for w in filtered_tokens]

# Helper function to map NLTK POS tags to WordNet POS tags for accurate
lemmatization
def get_wordnet_pos(treebank_tag):
    if treebank_tag.startswith('J'):
        return wordnet.ADJ
    elif treebank_tag.startswith('V'):
        return wordnet.VERB
    elif treebank_tag.startswith('N'):
        return wordnet.NOUN
    elif treebank_tag.startswith('R'):
        return wordnet.ADV
    return wordnet.NOUN

# 4. Lemmatization (using WordNetLemmatizer with POS tags)
lemmatizer = WordNetLemmatizer()
pos_tags = nltk.pos_tag(filtered_tokens)
lemmatized_words = [
```

```
    lemmatizer.lemmatize(w.lower(), get_wordnet_pos(tag))
    for w, tag in pos_tags
]

# Display Results
print("=" * 70)
print("1. ORIGINAL TEXT:")
print(text)
print("=" * 70)

print("\n2. TOKENS:")
print(tokens)

print("\n3. AFTER STOP-WORD REMOVAL:")
print(filtered_tokens)

print("\n4. STEMMED WORDS (Porter Stemmer):")
print(stemmed_words)

print("\n5. LEMMATIZED WORDS (WordNet Lemmatizer with POS):")
print(lemmatized_words)

print("=" * 70)
print("6. COMPARISON: STEMMING vs. LEMMATIZATION")
print("=" * 70)
print(f"{'Token':<15} | {'Stemmed Form':<15} | {'Lemmatized Form':<15}")
print("-" * 50)
for original, stem, lemma in zip(filtered_tokens, stemmed_words,
lemmatized_words):
    print(f"{original:<15} | {stem:<15} | {lemma:<15}")
```

output:

```
======================================================================
1. ORIGINAL TEXT:
The students are studying different subjects and they are enjoying their studies. The teachers are teaching the subjects very effectively.
======================================================================

2. TOKENS:
['The', 'students', 'are', 'studying', 'different', 'subjects', 'and', 'they', 'are', 'enjoying', 'their', 'studies', 'The', 'teachers', 'are', 'teaching', 'the', 'subjects', 'very', 'effecti

3. AFTER STOP-WORD REMOVAL:
['students', 'studying', 'different', 'subjects', 'enjoying', 'studies', 'teachers', 'teaching', 'subjects', 'effectively']

4. STEMMED WORDS (Porter Stemmer):
['student', 'studi', 'differ', 'subject', 'enjoy', 'studi', 'teacher', 'teach', 'subject', 'effect']

5. LEMMATIZED WORDS (WordNet Lemmatizer with POS):
['student', 'study', 'different', 'subject', 'enjoy', 'study', 'teacher', 'teach', 'subject', 'effectively']
======================================================================
6. COMPARISON: STEMMING vs. LEMMATIZATION
======================================================================
Token           | Stemmed Form    | Lemmatized Form
--------------------------------------------------
students        | student         | student
studying        | studi           | study
different       | differ          | different
subjects        | subject         | subject
enjoying        | enjoy           | enjoy
studies         | studi           | study
teachers        | teacher         | teacher
teaching        | teach           | teach
subjects        | subject         | subject
effectively     | effect          | effectively
```

==========

1. ORIGINAL TEXT:

The students are studying different subjects and they are enjoying their studies. The teachers are teaching the subjects very effectively.

==========

2. TOKENS:

['The', 'students', 'are', 'studying', 'different', 'subjects', 'and', 'they', 'are', 'enjoying', 'their', 'studies', 'The', 'teachers', 'are', 'teaching', 'the', 'subjects', 'very', 'effectively']

3. AFTER STOP-WORD REMOVAL:

['students', 'studying', 'different', 'subjects', 'enjoying', 'studies', 'teachers', 'teaching', 'subjects', 'effectively']

4. STEMMED WORDS (Porter Stemmer):

['student', 'studi', 'differ', 'subject', 'enjoy', 'studi', 'teacher', 'teach', 'subject', 'effect']

5. LEMMATIZED WORDS (WordNet Lemmatizer with POS):

['student', 'study', 'different', 'subject', 'enjoy', 'study', 'teacher', 'teach', 'subject', 'effectively']

==========

6. COMPARISON: STEMMING vs. LEMMATIZATION

==========

Token | Stemmed Form | Lemmatized Form

---

students | student | student

studying | studi | study

different | differ | different

subjects | subject | subject

enjoying	enjoy	enjoy
studies	studi	study
teachers	teacher	teacher
teaching	teach	teach
subjects	subject	subject
effectively	effect	effectively